(Please write your Enrollment Number) Enrollment No:

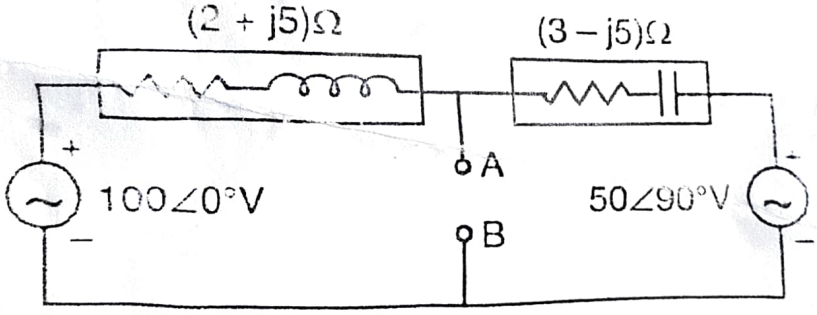
MID-TERM EXAMINATION

(Programme: B. Tech. ECE/ECE-AI) (Semester: II)

(March 2026) OFF LINE mode

Subject Code: BEC 204	Subject: Network Analysis and Synthesis
Time : 1 ½ hours	Maximum Marks :30

Note: Q1 is **compulsory. Attempt any two parts of Q2 and Q3**

Q1		**(2.5*4=10)**
	a) State Superposition Theorem.	
	b) What is the condition for maximum power transfer in a DC and in AC network?	
	c) At what time does the current reach 63.2% of its final value in an RL circuit?	
	d) Why does current in an inductor does not change instantaneously?	
	UNIT I	
Q2	**Attempt any two parts**	**(5*2=10)**
	a) State Thevenin's Theorem and mention its advantages and limitations.	
	b) Find the Thevenin's resistance of the network shown in Fg-1	
	 Fig:1	
	c) Find the Thevenin's Voltage of the network shown in Fig-1	
	UNIT II	
Q3	**Attempt any two parts**	**(5*2=10)**
	A series RL circuit has: Resistance R=10Ω Inductance L=2H, Applied DC voltage V=50V Calculate: a. Time constant of the circuit	
	b) Expression for current growth i(t)	
	c) Current at t=0.2 seconds	